

Project plan

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The purpose of this master thesis is to formulate string-string interactions as a scattering problem, including developing quantum shoving of strings and extracting a cross-section for the scattering.

The project can be divided into 4 phases:

Phase 1: Establish and understand full analytical and numerical control of a two-site model. Also, explore parameters and robustness.

Phase 2: Extend the two-site model to a spatially resolved one-dimensional lattice and study the time-dependent Schrödinger evolution, where analytical solutions are not available.

Phase 3: Formulate string-string interactions as an effective scattering problem and extract cross sections.

Phase 4: Explore extensions of the baseline model depending on progress and results in previous phases. (Phase 4 is not included in the Time plan as it depends on how much time is left until deadline)

Timeplan

The time plan is presented below. The first weeks are detailed and it is clear when things need to be done. For the rest of the weeks, it is difficult to guess how long each task will take. The timeplan will most likely be improved after the first 3-4 weeks. The report will be written throughout the weeks, where most of the writing is expected to be done in Phase 2 and 3.

Timeline	Phase 1	Phase 2	Phase 3
Week 1	Understand the two-site system and reproduce derivations given by supervisor.		
Week 2	Implement time evolution for the two-site Hamiltonian using a numerical method (Crank-Nicholson). Compute transition probability and $\langle T \rangle(t)$. Compare numerical results to analytical.		
Week 3	Perform systematic scans over V/J and J . Identify regions where transition is strongly suppressed. Quantify parameters transition probability depends on.		
Weeks 4-8		Understand and implement a 1D lattice Schrödinger solver. Validate solver with known cases. Compute reflection, transmission and momentum-transfer observables. Have a verified numerical solver (convergence test!). Have time-dependent wavefunction and momentum-space distributions.	
Weeks 9-17			Define a scattering setup in 1D (maybe 2D). Identify transmission/reflection coefficients and scattering cross sections. Derive analytical estimates using WKB methods. Have parameter-dependent scattering probabilities. Have quantitative comparison between WKB and numerical methods.